University of Central Punjab

Faculty of Information Technology

Introduction to Computing - Lab 03

Topic

Hello World Program, Arithmetic Expressions, Variables, and Interactive Programs (Taking Input from Console). Flowcharts for simple programs.

Objective

- Write, compile and run a simple C++ program and understand its execution flow.
- Use cout to print text, numbers and arithmetic expressions.
- Evaluate arithmetic expressions using operator precedence and associativity.
- Declare, initialize and use variables of different types.
- Take input from the user using cin and understand how white space is handled.
- Represent a simple program as a flowchart and write a program from a flowchart.

Outcomes

1. Write a Hello World style program using cout, comments and escape sequences.
2. Identify syntax errors and explain the difference between syntax and semantics.
3. Evaluate arithmetic expressions involving +, -, *, / and % correctly, including integer division.
4. Declare and initialize variables, follow identifier rules, and use assignment statements.
5. Write interactive programs using cin, and draw and read flowcharts to solve simple problems.

Content

The following sections will be covered during this lab session.

1. **Hello World Program** will cover execution flow, the syntax of cout with text and numbers, comments, syntax errors, and syntax vs. semantics.
2. **Arithmetic Expressions** will cover the operators (+, -, *, /, %), operator precedence and associativity, and printing the value of an expression.
3. **Variables** will cover the purpose of variables, types, declaration and initialization, valid identifiers, and assignment statements.
4. **Interactive Programs** will cover cin, how white space (space, tab, newline) is treated, and problem solving using variables.
5. **Flowcharts** will cover the basic symbols, reading a flowchart, finding mistakes in it, and drawing your own.

Instructor: Maria Faizi

Instructions

- Indent your code properly.
- Use meaningful variable names.
- Use meaningful prompt lines/labels for all input/output.
- Your name and roll no must be mentioned in the start of your file as comment. For example, **//L1F23BSCS0000 – Ahmad**
- You may use only what has been covered in this lab (cout, cin, variables and arithmetic expressions). Do not use if, loops or functions in your programs.
- Every task has an Expected Output. Your output must match it for the given inputs.
- Read the Introduction to Flowcharts below before starting Task 06.
- You are required to complete the following tasks in lab timings (3 hours). The tasks get a little harder as you go.

Introduction to Flowcharts

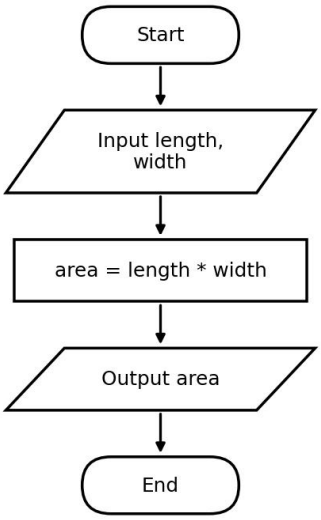
A flowchart is a picture of the steps of a program. Every step is written inside a symbol, and arrows show the order in which the steps happen. Drawing the flowchart first helps you plan the program before you write any code. These are all the symbols you need:

Symbol	Name	Used for	Example
	Terminator (oval)	Start and End of the program	Start, End
	Input / Output (parallelogram)	Taking input or showing output	Input price, quantity
	Process (rectangle)	A calculation or an assignment	total = price * quantity
	Decision (diamond)	A yes / no question. It has two exits, marked Yes and No	marks >= 60 ?
	Connector (circle)	Joins two parts of a big flowchart, or continues it on another page	A
	Flow line (arrow)	Shows which step comes next	Top to bottom

Rules

- Every flowchart has one Start and one End.
- Steps are read from top to bottom by following the arrows.
- Write one step in one symbol, and use the same variable names that you will use in the program.
- A decision symbol has one arrow coming in and two going out (Yes and No). Only one of the two paths is followed.

Example 1: A carpet shop needs the area of a room (no decision)



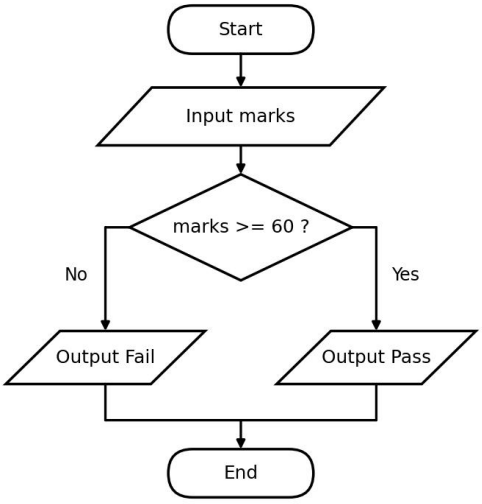
The same flowchart written as a program:

```
int length, width, area;

cin >> length >> width;
area = length * width;
cout << area;
```

Each box of the flowchart becomes one line of the program.

Example 2: A teacher decides Pass or Fail (with a decision)



How to read it:

1. The program takes the marks as input.

2. The diamond asks: is marks >= 60 ?

3. If the answer is Yes, the right arrow is followed and Pass is printed.

4. If the answer is No, the left arrow is followed and Fail is printed.

5. Both paths join again and the program ends.

For marks = 75 the output is Pass, and for marks = 45 the output is Fail.
In C++ a decision is written using if. You will learn it in the next lab. For now, you only have to read and draw it.

Task 01 – Welcome Banner (Hello World and cout)

Estimated time: 10 minutes

> *The university is organising an Orientation Day. You have been asked to write a program that prints a welcome banner on the screen of the registration desk.*

Requirements

1. Start the file with your name and roll no as a comment, and write a short // comment above each part of the program.
2. Print Hello World! on the first line.
3. Using only one cout statement and \n inside the text, print these three lines:

```
University of Central Punjab
Faculty of Information Technology
Orientation Day 2026
```

4. Print the line Welcome! using endl.
5. Print the numbers 2026 and 45, and the expressions 45 * 3 and 1000 + 500 - 250, each on its own line.
6. Print Total fee: 54000 using one cout statement, where 54000 comes from the expression 45000 + 9000.

Expected Output

```
Hello World!
University of Central Punjab
Faculty of Information Technology
Orientation Day 2026
Welcome!
2026
45
135
1250
Total fee: 54000
```

Observation

Remove the semicolon from one line, and write Cout instead of cout in another. Note both compiler messages as comments, then fix them.

Task 02 – Debug the Code (Syntax Errors vs. Semantics)

Estimated time: 15 minutes

> *A junior student has sent you his first program. It does not compile, and even after it compiles, one answer looks wrong. Help him.*

Part A – Syntax errors

The following program has four syntax errors. Type it exactly as given, read the compiler messages, and write each error as a comment. Then correct the program.

```
#include <iostream>
int main()
{
    cout << "Orientation Day" << endl
    cout << "Total students: " 450 << endl;
    cout << "Seats left: " << 500 - 450 << endl;
    cout << 'Have a nice day << endl;
    return 0;
}
```

Part B – Semantic error

This line compiles without any error, but the answer is wrong. The average of 60, 70 and 80 should be 70.

```
cout << "Average: " << 60 + 70 + 80 / 3 << endl;
```

1. Run it. What is printed? Fix the line using brackets.
2. In one line each, write in a comment what a syntax error is and what a semantic error is. Which one does the compiler catch?

Expected Output

```
Part A (corrected program):
Orientation Day
Total students: 450
Seats left: 50
Have a nice day

Part B:
Average: 156     (wrong, before the fix)
Average: 70      (correct, after the fix)
```

Task 03 – Expression Quiz (Precedence and Associativity)

Estimated time: 15 minutes

> *Your teacher gives a quiz in which you must evaluate expressions without a calculator. Check your answers using the compiler.*

Requirements

1. BEFORE running anything, fill the Your Prediction column of the table on paper.
2. Print every expression using one cout statement, for example: cout << "Expression 1: " << 10 + 5 * 2 << endl;
3. Write the actual output next to your prediction as a comment at the end of your file.

No.	Expression	Your Prediction	Actual Output
1	10 + 5 * 2		
2	(10 + 5) * 2		
3	20 - 5 - 3		
4	100 / 7		
5	100 % 7		
6	2 * 3 + 8 / 4 - 1		

4. For **expression 3**, write in a comment whether the operators are evaluated left-to-right or right-to-left.
5. Add brackets to **expression 6** so that its result becomes 8.

Task 04 – Admission Form (Variables and Types)

Estimated time: 20 minutes

> *A new student is admitted. The admission office wants to store the student's data in variables, update the fee record, and find out how much fee is still due.*

Requirements

1. Declare and initialize variables of a suitable type: rollNo (1045), semester (1), section ('B'), cgpa (3.5), totalFee (45000) and feePaid (20000).
2. Display all the values with proper labels.
3. Move the student to the next semester using semester = semester + 1; and add the second installment of 5000 to feePaid using an assignment statement.
4. Calculate and display remainingFee **(totalFee - feePaid)** and the percentage of the fee paid **(feePaid * 100 / totalFee)**.
5. In a comment, mark each name as a valid or invalid identifier: 2ndYear, student_name, total-marks, _fee, Roll No, marks2.

Expected Output

```
Roll No: 1045
Semester: 1
Section: B
CGPA: 3.5
Total Fee: 45000
Fee Paid: 20000
Semester: 2
Fee Paid: 25000
Remaining Fee: 20000
Percentage Paid: 55
```

Observation (write the answer as a comment)

Declare int x; without any initial value and print it. What do you see? Is it safe to rely on it? (Expected: some unpredictable number, so it is never safe.)

Task 05 – Fee Calculator (cin and White Space)

Estimated time: 20 minutes

> *At the registration desk, the student types in the number of credit hours and the program calculates the semester fee.*

Requirements

1. Declare int variables age, creditHours and feePerCreditHour.
2. Using cin and a meaningful prompt for each, read the age, the credit hours and the fee per credit hour.
3. Calculate and print the total fee (creditHours * feePerCreditHour), and the fee after a discount of Rs 2000.
4. Print the year of birth (assume the current year is 2026), and the year in which the student will turn 25.

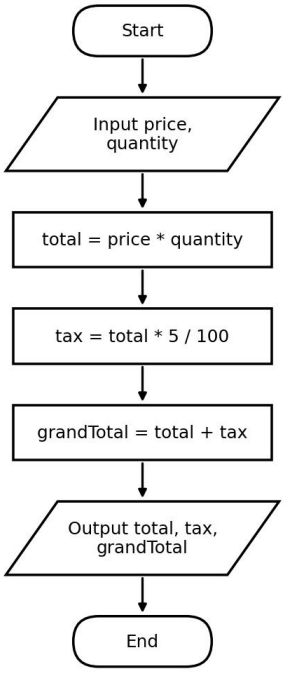
Experiments (write what you observed as comments)

1. Run the program and type all three values on one line separated by spaces (19 15 8000). Then type each value on a separate line, and then separated by tabs. Is the result different? Why?
2. Type the age as 19.8. What is stored in age, and what happens to the next value? Explain why.

Task 06 – Bookshop Bill (Flowchart to Program)

Estimated time: 20 minutes

> *A bookshop wants a program that prints the bill of a customer. The manager has drawn the flowchart of the program, shown below. You have to turn it into a C++ program.*



Requirements

1. Write the program exactly as the flowchart shows, with the same variable names (all int). Read the values using cin.
2. Dry run: for price = 250 and quantity = 3, fill in the table below on paper BEFORE running the program, and then compare with the actual output.

Variable	Your Prediction	Actual Output
total		
tax		
grandTotal		

3. Run the program again with price = 99 and quantity = 7. A calculator gives a tax of 34.65. What does your program print, and why is it different? Answer in a comment.

Task 07 – Weather App (Draw the Flowchart, then Code)

Estimated time: 15 minutes

> *A weather app shows the temperature in Celsius, but tourists want to see it in Fahrenheit as well. The formula is: Fahrenheit = Celsius * 9 / 5 + 32.*

Requirements

1. FIRST, draw the flowchart of the program using the correct symbols (Start, Input, Process, Output, End). Show it to your instructor before you start coding.
2. Write the program. Take the temperature in Celsius from the user using cin, calculate the Fahrenheit value, and print both with proper labels.
3. Test with 0, 100 and 37. A calculator gives 98.6 for 37. What does your program print, and why? Answer in a comment.

Expected Output (the numbers typed by the user are shown after the prompts)

```
Enter temperature in Celsius: 100
Temperature in Fahrenheit: 212

Celsius 0   gives Fahrenheit 32
Celsius 37  gives Fahrenheit 98   (the .6 is lost, integer division)
```

Task 8 – Cinema Ticket (Reading a Flowchart with a Decision)

Estimated time: 15 minutes

> *A cinema gives a half ticket to children below 12 years and a full ticket to everyone else. The counter staff use a flowchart to decide which ticket to print. There is no coding in this task: you only read and draw flowchart.*

Requirements

1. Draw a complete flowchart for a voting eligibility program. Input age. If age >= 18, output Eligible to vote, otherwise output Not eligible to vote. Use all the required flowchart symbols, including the decision symbol.
2. Dry run the flowchart for the ages 8, 12 and 30. Write which ticket is printed for each age.

No.	Age	Condition (Age<12)	Ticket Printed
1			
2			
3			

Note:

- Write comments in your program, and use meaningful variable names.
- You are not allowed to use any built-in functions.